Relaytic-AML: A Local-First Agentic Evaluation Lab for Financial-Crime Machine Learning

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Abstract

Anti-money laundering (AML) machine-learning experiments are hard to audit when private data, temporal validity, graph provenance, leakage controls, review capacity, and public claims are managed in separate tools. Relaytic-AML is a local-first evaluation lab in which role-scoped agents produce auditable evidence cells and deterministic gates route those cells into admissible paper uses. The current evidence pack reports PaySim synthetic temporal-fraud PR-AUC 0.6388, Elliptic temporal graph-feature PR-AUC 0.6688, and Elliptic2 context PR-AUC 0.9432 +/- 0.0009, with a recorded RevClassifyDS reference of 0.9740. The paper contributes an evaluation and governance substrate for financial-crime ML, with detector claims kept within the evidence boundary.

1 Introduction

AML modeling is a rare-event, high-stakes evaluation problem. The input may be a stream of mobile-money transfers, card events, account activity, wire messages, customer profiles, or blockchain transactions. Suspicion rarely appears in one row. It often appears as a temporal or network pattern: fast movement of funds after receipt, repeated transfers through related accounts, structured amounts, new counterparties receiving many payments, activity that does not match a customer's profile, or cryptocurrency behavior that involves risky services and geography. Regulatory and typology material from FATF, FinCEN, and FFIEC describes this kind of pattern-based reasoning in operational language [Financial Action Task Force, 2020, Financial Crimes Enforcement Network, 2026, Federal Financial Institutions Examination Council, 2026].

A useful score is not enough. A reader needs to know which data stayed local, which columns were available at decision time, how time or graph boundaries were split, whether model selection touched the test surface, what review budget was assumed, and what interpretation the evidence can support. These questions become sharper when large language models (LLMs) or coding agents assist the research workflow, because fluent explanations can drift away from the artifact record unless the system is designed to fail closed.

Relaytic began as a general local-first inference-engineering lab. Relaytic-AML is the financial-crime edition used here to test whether that architecture can support governed AML experimentation. It is a set of cooperating agents and deterministic harnesses around a local artifact store. The guide helps a user or another agent understand where the run is. The scout checks source posture, schema, leakage risk, and split feasibility. The strategist turns the objective into a task contract. The scientist challenges baselines, ablations, and budget choices. The builder executes bounded runs. Reviewers reconstruct traces. Release governors lint claims, figures, tables, source packages, and public wording against the evidence record.

The paper is therefore a systems and methodology paper. The benchmark rows matter because they exercise the architecture under temporal, graph, operating-point, and claim-governance pressure. The central question is whether a local evaluation lab can keep evidence, agent assistance, privacy boundaries, and publishable claims aligned without turning proxy evidence into detector-superiority claims.

The work is organized around four research questions:

- **RQ1:** Can Relaytic-AML produce reproducible evidence cells for AML-style temporal and graph tasks?
- **RQ2:** Can it prevent leakage-prone or unsupported claims from being promoted?
- **RQ3:** Can the local artifact record support rowless handoff to external agents while preserving provenance?
- **RQ4:** Do the benchmark rows demonstrate useful, bounded detector evidence under explicit split and budget contracts?

This paper makes four contributions. First, it presents a workspace-backed, role-scoped agent runtime for AML evaluation. Second, it defines an evidence-cell schema tying each metric to dataset, split, command, artifact, budget, leakage posture, and operating point. Third, it contributes a release and interpretation-routing harness that turns local artifacts into tables, figures, source packages, and manuscript claims while recording the evidence needed for stronger future uses. Fourth, it gives public demonstrations on PaySim, Elliptic, and Elliptic2 reference rows under explicit evidence roles.

2 Related Work

The AML benchmark landscape is moving toward larger, more realistic, and more graph-native settings. PaySim remains useful as a synthetic mobile-money fraud simulator for temporal proxy experiments [Lopez-Rojas et al., 2016]. Elliptic introduced a public Bitcoin transaction graph with anonymized node features and temporal labels [Weber et al., 2019]. Elliptic2 and RevTrack/RevClassify shifted attention to suspicious subgraphs and richer blockchain context [Bellei et al., 2024, Song et al., 2024]. Recent work such as TransXion, LineMVGNN, quasi-temporal graph extraction, BlazingAML, and continual graph-learning studies shows that the research frontier increasingly treats AML as dynamic graph and systems work rather than static tabular classification [Chen et al., 2026, Poon et al., 2026, Tariq and Hassani, 2026, Ye et al., 2026, Deprez et al., 2025].

Detector papers such as TransXion, BlazingAML, LineMVGNN, and RevClassifyDS push the model frontier. Relaytic-AML sits one layer around that work: it asks how experiments should be governed when data is local or licensed, the task may be temporal or graph-based, analyst review capacity matters, and an agent-assisted workflow must still produce evidence that a skeptical reviewer can audit. That places the system near dataset documentation, model reporting, reproducibility practice, experiment tracking, and governance work [Gebru et al., 2021, Mitchell et al., 2019, Pineau et al., 2021].

The system also differs from adjacent evaluation artifacts. Model cards explain a trained model, but they do not usually bind every number to a command, split, artifact field, and admissible-use record. Datasheets describe data, but they do not run the model-search and release gates. Reproducibility checklists improve reporting, but they are often static forms rather than executable evidence. MLOps experiment trackers preserve runs, but they are not designed to govern public scientific claims about local, licensed, or privacy-sensitive AML data. Agent benchmarks evaluate agents, while Relaytic-AML uses agents inside an evaluation lab and then tests whether the lab keeps the agents attached to artifacts. This is the systems contribution: evidence, agents, local privacy, and publishable wording are coupled in one deterministic release path.

Recent agent-evaluation work shows that language-model systems can produce persuasive research artifacts while still failing on reproduction, validation, or expert-level judgment [Chen et al., 2025, Starace et al., 2025, Wijk et al., 2025]. Skill- and tool-using agents make that opportunity larger and the governance problem sharper [Yang et al., 2026]. Relaytic-AML responds by making model scores, public claims, handoff packets, and paper assets downstream of local artifacts rather than downstream of a conversation transcript.

3 System Overview

Relaytic-AML is built around one authority rule: the workspace owns the truth. Raw data, licensed benchmark files, run summaries, traces, metric cells, model outputs, tables, figures, and release reports live on disk. Agents may explain, propose, and repair, but their proposals only become evidence when they are materialized as artifacts another human or agent can inspect.

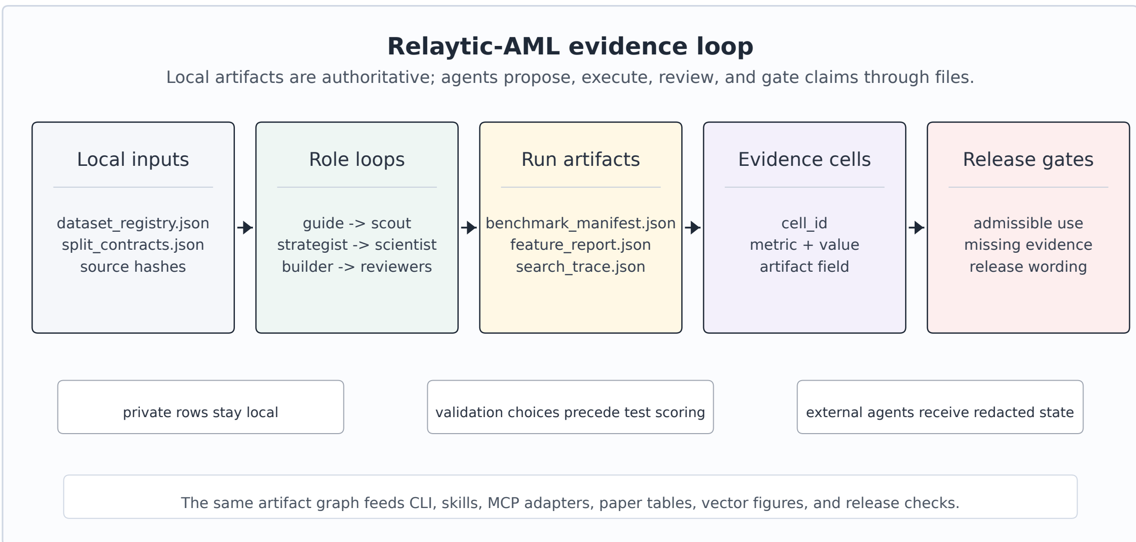

Figure 1: Relaytic-AML local-first architecture: local data and artifacts flow through role-scoped agents into evidence cells, interpretation gates, and paper/release/handoff surfaces.

Figure 1 summarizes the local evidence loop. Dataset registries and split contracts enter the role-scoped agent runtime. Candidate runs write benchmark manifests, search traces, feature reports, and metric cells. Claim gates read those cells together with release audits and emit only the interpretations that the evidence supports. The same contract feeds the command-line interface, project skills, OpenClaw-style handoff, Claude/Codex skill files, and Model Context

Protocol (MCP) adapters.

The external-agent path is rowless by default. Relaytic can export current state, next-action options, artifact shortlists, and safe commands for another model without sending raw transaction rows, secrets, or private local paths. A local LLM can optionally help phrase guidance, but the deterministic guide and evidence artifacts remain the source of truth. That separation matters for private AML work: outside intelligence may help navigate the run, but data residency and claim provenance stay local unless an operator deliberately changes policy.

4 Evidence Cell and Claim-Gate Design

An evidence cell is the unit that makes a paper number auditable. It is not just a metric value. It records the dataset, split, command, artifact field, model or feature budget, leakage posture, and operating point. Interpretation is deliberately stored in a separate gate output: the cell says what happened, and the gate says how that fact may be used.

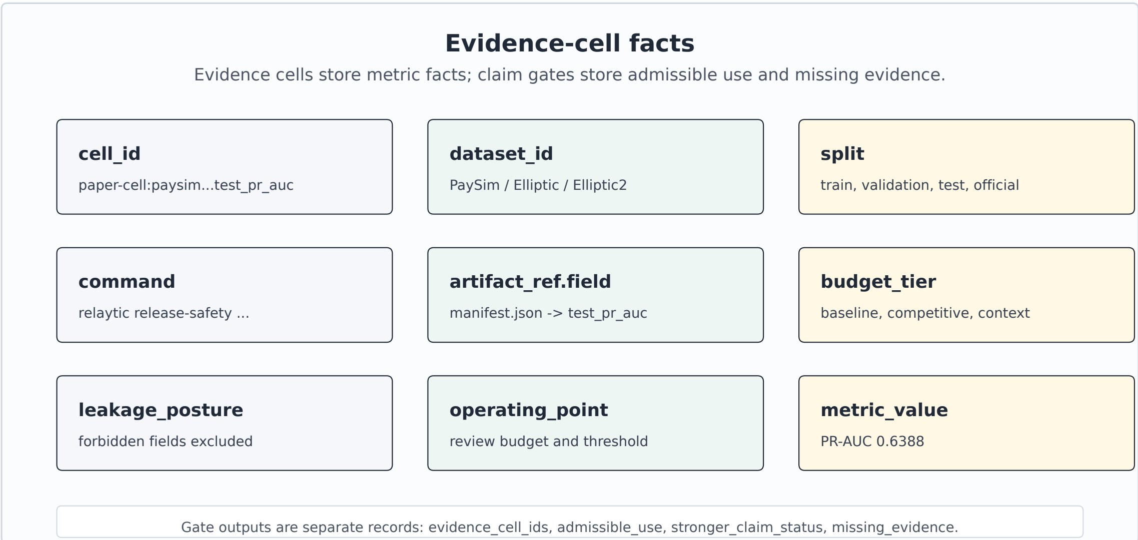

Figure 2: Evidence-cell schema: every reported number carries dataset, split, command, artifact, budget, leakage posture, operating point, metric, and value; interpretation is stored separately.

Table 1 uses compact publication aliases for readability; the full machine metric-cell identifiers are preserved in the metric-cell audit artifact and generated table comments.

Table 1. Representative evidence cells.

ID	Dataset	Metric	Value	Split	Artifact	Evidence role
PS-PR	PaySim	test PR-AUC	0.6388	temporal test	PaySim run; manifest	bounded demonstration
PS-P@B	PaySim	precision at review budget	0.7033	temporal test	PaySim run; manifest	bounded demonstration
EL-PR	Elliptic	test PR-AUC	0.6688	graph-time test	graph run; feature table	graph-feature evidence
EL-P@B	Elliptic	precision at review budget	1.0000	graph-time test	graph run; feature table	graph-feature evidence
E2-PRm	Elliptic2	official test PR-AUC mean	0.9432	official test	E2 run; scorecard	external reference/context
E2-ref	Elliptic2	published reference PR-AUC	0.9740	reported ref.	E2 run; scorecard	external reference/context

A representative record is compact enough to audit directly. The public table uses the alias `PS-PR`, while the underlying artifact keeps the longer machine identifier. The example separates the factual evidence cell from the gate output that names admissible use and evidence needed for a stronger interpretation.

```
{
  "evidence_cell": {
    "cell_id": "PS-PR", "dataset_id": "paysim_temporal_transaction_fraud",
    "split": "temporal test", "metric": "test_pr_auc", "value": 0.6388,
    "artifact_ref": "paper_metric_cell_audit.json:test_pr_auc",
    "budget": "competitive", "leakage_posture": "balance/raw IDs excluded",
    "operating_point": "ranking metric"
  },
  "claim_gate_output": {
    "evidence_cell_ids": ["PS-PR"], "admissible_use": "bounded PaySim proxy",
    "stronger_claim_status": "requires external holdout",
    "missing_evidence": ["partner holdout", "incumbent queue study"]
  }
}
```

Algorithm 1 Evidence-cell creation

Input: dataset registry D, split contract S, candidate budget B, run artifacts A
Output: evidence cell c with factual provenance
Freeze source posture, license posture, task target, and split contract.
Derive only features allowed by S and record excluded leakage fields.
Run baseline candidates under the declared baseline budget.
Run stronger candidates only within B and select on validation evidence.
Evaluate the selected row once on the fixed test surface.
Write c = dataset, split, command, artifact field, metric, value, budget, leakage posture, and operating point.
Hand c to the claim gate before it appears in tables, figures, or release text.

The claim gate is the second half of the design. It is deliberately conservative. If the evidence cell is incomplete, if a split is leakage-prone, if a metric is only a proxy, or if a stronger interpretation needs a different dataset or study, the gate preserves the evidence and routes the stronger use

Algorithm 2 Claim-gate validation

Input: public claim q, evidence cells C, gates G, limitations L
Output: admissible wording and evidence-needs record
Resolve every evidence cell named by q and require dataset, split, command, artifact, budget, and leakage fields.
Compare the strength of q with source posture, split validity, metric scope, and benchmark role.
If q is exactly supported, emit the bounded wording and the evidence-cell identifiers.
If q is stronger than C and G permit, record the stronger-claim status and gate reason.
Attach the missing evidence needed to make q testable in future work.
Route current evidence to its admissible paper use and keep stronger uses out of headline wording.

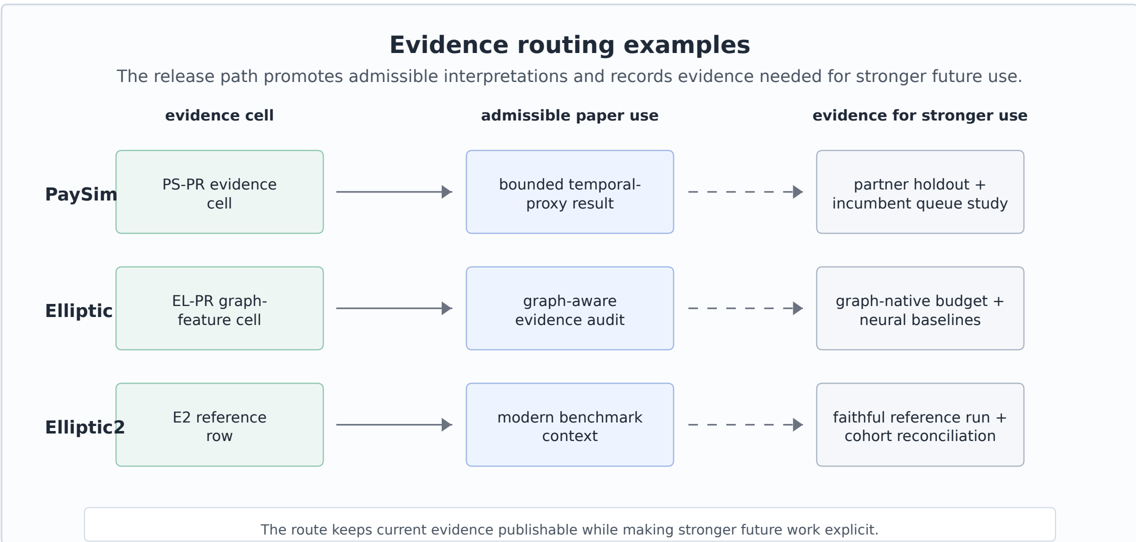

Figure 3: Evidence routing examples: current cells map to admissible paper uses and to evidence needed for stronger future interpretations.

to an evidence-needs record. This is a mechanism, not a disclaimer: it changes what the paper generator and public release surfaces are allowed to say.

Figure 3 gives concrete routing behavior. A PaySim row becomes a bounded temporal-proxy demonstration, an Elliptic row becomes graph-feature evidence with temporal provenance, and an Elliptic2 row becomes external benchmark context. The same records also specify what evidence would be needed before stronger future uses could be made.

5 Experimental Protocol

The experiments test Relaytic-AML as an evaluation lab. The datasets are separated by evidence role. PaySim is the main empirical demonstration because it is fully local and supports a controlled temporal proxy workflow. Elliptic tests temporal graph provenance and feature-view discipline. Elliptic2 tests whether the system can keep a modern external reference row visible without overstating its role.

Table 2a. Dataset scale and split contracts.

Dataset	Task	Scale and positives	Train / validation / test	Split rule	Source hash
PaySim synthetic mobile-money transaction fraud	Synthetic temporal transaction fraud	6,362,620 rows/nodes; 8213 positives	train: 6,010,937, pos 0.08%; validation: 228,103, pos 0.68%; test: 123,580, pos 1.34%	time step	sha256 prefix 16910f90577b
Elliptic Bitcoin transaction graph	Temporal Bitcoin graph node classification	203,769 rows/nodes; 4545 positives	train: 26,381, pos 10.88%; validation: 8,999, pos 11.53%; test: 11,184, pos 5.69%	graph time	sha256 prefix 93e2e7b2405c plus 2 files
Elliptic2 large Bitcoin subgraph AML dataset	Suspicious-versus-licit subgraph context	122,000 labeled subgraphs; context row	val 11,059, test 11,105; test positives 272	fixed partition	not bundled

Table 2b. Feature and metric policy.

Track	Allowed feature policy	Forbidden or gated inputs	Primary metrics	Evidence role
PaySim	26 decision-time features; balance columns excluded	balance columns, raw account IDs, simulator flags, random row shuffles	PR-AUC, P@k, R@budget	bounded demonstration
Elliptic	structural same snapshot only; source provided flattened features; source features plus structural snapshot	future-to-train edges and random node splits across time	PR-AUC, P@k, fixed-FPR recall	graph-feature evidence
Elliptic2	pooled moments counts; raw subgraphs local-gated	claim use gated by reference parity and local data availability	PR-AUC, P@k, fixed-FPR recall	external reference/context

Tables 2a and 2b record the context a reader needs before interpreting any metric. The positive rates show why precision-recall area under the curve (PR-AUC) is the primary score. These are rare-event tasks where receiver-operating-characteristic area under the curve can look strong while the review queue remains poor. The split rules are equally important: PaySim is split by chronological simulator step, Elliptic by time-step graph windows, and Elliptic2 by the official/context partitions recorded in the local evidence pack.

Table 3. Model families and search budgets.

Track	Families	Features	Search budget	Evidence role
PaySim	tree and boosting candidates; Extra Trees selected	amount, type, time, shifted destination history	14 probes; 5 finalists; seeds 42	validation PR-AUC; one fixed test
Elliptic	tree/boosting baselines; LightGBM selected	source node features plus same-step graph statistics	16 trials; seeds 42	graph-feature evidence row
Elliptic2	LightGBM context row	348 pooled subgraph moments/counts	3 repeated seeds: 11, 42, 73	external reference row

Table 3 records the modeling effort without presenting the paper as a hyperparameter leaderboard. PaySim uses a two-stage competitive budget: probes over a seeded train-only sample, five full-training finalists, validation-only calibration, and one fixed test evaluation for the selected finalist. Elliptic uses a graph-feature budget over source-provided anonymized features, same-snapshot structural features, and their combination. Elliptic2 uses repeated pooled-moment LightGBM context rows with seeds 11, 42, and 73 as an external-reference stress test for the release machinery.

The feature policy is strictest for PaySim because simulator balance columns can leak post-event state. Relaytic excludes those balance fields, raw origin and destination identifiers as model features, and simulator flags. It allows row-local amount, type, and time features, train-only thresholds, and destination history shifted before each step. The isolated contribution of destination-history features has not been measured as a separate test row, so it is not reported as a main result.

6 Results

Table 4. PaySim modeling path.

Stage	Model/contract	Selection evidence	Final test evidence	Role
P4 reference	SGD logistic baseline	source-safe starting point	0.2159	reference row
P6 baseline	Extra Trees baseline	leakage-safe feature set	0.3313	baseline row
Competitive search	XGBoost probe	26 allowed features; best probe validation PR-AUC 0.5944	not evaluated on test	candidate screening
Final model	Extra Trees with Platt calibration	validation-selected finalist; validation-only calibration	0.6388	bounded demonstration

PaySim is the clearest modeling result in the current evidence pack. The earliest reference row was PR-AUC 0.2159. The later leakage-safe baseline improved to 0.3313, and the competitive selected Extra Trees model reached fixed-test PR-AUC 0.6388 and ROC-AUC 0.9683. The improvement is meaningful inside the synthetic temporal-fraud contract because it follows a visible sequence: balance fields were excluded, prior-step destination history was added without raw account encoding, candidates were selected by validation evidence, and calibration and thresholding used validation-only partitions. The admissible interpretation is precise: Relaytic-AML produced a stronger, leakage-audited PaySim temporal-proxy row under a declared budget.

The review-budget metrics sharpen the interpretation. At the selected PaySim review budget, precision is 0.7033 and recall is 0.4716. The top of the queue is much richer than prevalence, but a large share of fraud remains outside the reviewed set. That is a useful operating result for an evaluation lab because it connects ranking quality to analyst capacity instead of treating PR-AUC as the whole story.

Elliptic is a different kind of evidence. The validation-selected source-plus-structural LightGBM row reports test PR-AUC 0.6688, with review-budget precision 1.0000 and recall 0.0566. The result supports temporal graph-feature provenance and operating-point reporting. It also reveals a limitation: the current graph-structure-only floor is weak, and the final row is heavily influenced by source-provided anonymized features. Relaytic's contribution here is the graph-aware evidence path: feature provenance, temporal splits, operating-point metrics, and interpretation routing are made auditable together.

Elliptic2 is intentionally framed as an external reference row. The repeated official-partition context row reports PR-AUC 0.9432 +/- 0.0009, and the content-hash robustness partition reports mean PR-AUC 0.9297. Those are strong absolute values, and the recorded RevClassifyDS reference of 0.9740 gives the reader a useful frontier marker. Relaytic's evidence role is to keep that modern benchmark context visible while recording the cohort and reference-execution evidence needed for stronger future comparison.

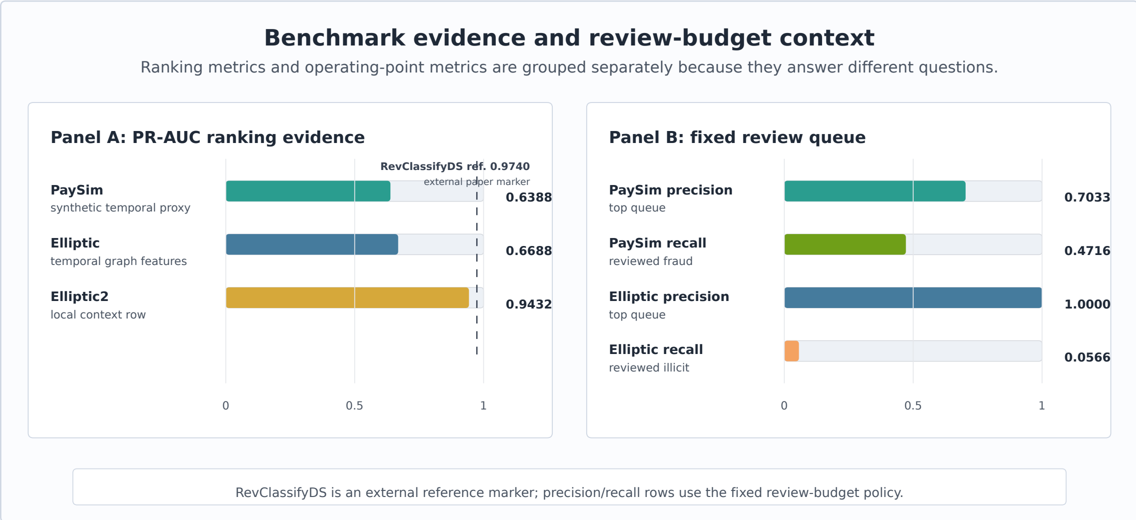

Figure 4: Benchmark and review-budget evidence: PR-AUC is shown beside precision and recall at the bounded review queue instead of being interpreted alone.

Figure 4 separates ranking metrics from operating-point metrics. PR-AUC summarizes ranking quality under rare-event imbalance. Precision and recall at the selected review budget describe what the top of an analyst queue would contain under the paper's fixed policy. Keeping those views together, but visually separated, is important because a useful top queue can still leave many positives unreviewed.

7 System Evaluation

The system claim is evaluated through deterministic reader and agent tasks. These checks ask whether a reviewer can navigate from the README to the paper evidence, trace PaySim metric provenance, compare baseline and competitive budgets, keep Elliptic2 as context rather than a contribution, export rowless state for an external agent, recover an interrupted run, and block over-strong public claims. This is narrower than a human-subject usability study, but it directly tests the infrastructure claim made by the paper.

Table 5. System audit matrix.

Check	Command or test	Evidence	Pass criterion	Observed result
Metric provenance	metric provenance	PaySim selected PR-AUC cell	required fields present	pass; fields 13/13 in metric-cell audit
Budget comparison	budget comparability	PaySim baseline and competitive cells	same dataset, split doctrine, and metric	pass; same split/metric; PR-AUC 0.3313 -> 0.6388
PaySim interpretation	interpretation-route check	PaySim publishability row	admissible use present; headline=false	pass; admissible use present; hard/headline=false
Elliptic2 reference role	reference role check	Elliptic2 publishability row	reference role visible; parity evidence required	pass; reference role visible; parity evidence required
Rowless handoff	handoff recovery	agent handoff report	state, tools, next action; no rows	pass; raw rows=false; redactions=8; blocked fields=6
Interrupted recovery	no-lost-user recovery	guide recovery report	stage, shortlist, next action exposed	pass; partial run; missing=8; actions=6
Stronger-use routing	routing cases	evidence-needs case studies	missing evidence recorded	pass; case studies record missing evidence

Table 5 reports the audit matrix behind the system claim. Each row names a behavior, a deterministic check, a pass criterion, and an observed signal: 13 of 13 provenance fields are present for the PaySim metric cell, baseline and competitive budgets are comparable under the same contract, Elliptic2 remains in its reference role, rowless handoff exposes no raw rows, and interrupted-run recovery surfaces state, missing evidence, and next actions.

Table 6. Evidence routing examples.

Stronger future use	Current admissible use	Evidence needed
Real-bank deployment study	bounded PaySim temporal-proxy demonstration	Partner or bank-approved holdout, incumbent comparison, analyst-review protocol, and legal release gate.
Elliptic2 reference-method comparison	external RevClassifyDS reference marker plus local context row	Faithful reference execution, cohort reconciliation, resource budget, and repeated parity report.
Graph-native detector release	Elliptic temporal graph-feature evidence path	Graph-native release budget, neural baselines, repeated seeds, and graph-specific ablations.

Table 6 shows how stronger future uses are handled. Rather than letting narrative claims drift beyond the artifacts, the gate records the current admissible use and the evidence that would be needed before the stronger interpretation could be made.

Table 7. Rowless handoff and interrupted-run recovery examples.

Scenario	Input state	Exported fields	Redacted fields	Observed signal
External-agent handoff	partial run with available guide state	state summary, action options, starter questions, tool contract, artifact shortlist	raw transaction rows, credentials, private paths, raw source files	raw_rows=False; redactions=8; blocked_fields=6
Safe next action	external model asked what to do next	six next actions, six starter questions, command options	unredacted local paths and data rows	actions=6; starter_questions=6
Interrupted-run recovery	operator returns to partial run without artifact literacy	current state, missing evidence count, canonical artifact shortlist, context-export command	raw benchmark data and private machine paths	state=partial_run; missing=8; actions=6

Table 7 gives the practical external-agent story. A second model can receive state, commands, artifacts, and starter questions, while raw rows remain redacted together with private machine paths. The same mechanism helps an inexperienced or interrupted user recover the next action without knowing which internal artifact to inspect first.

8 Limitations and Threats to Validity

PaySim is synthetic. It is useful for controlled temporal fraud experiments, but it is not evidence of bank-scale AML superiority. The simulator has known simplifications, and the current result should be interpreted as a leakage-audited proxy result. The destination-history feature contract is present, but the isolated destination-history ablation is not in the current evidence pack, so no separate result is claimed for that feature family.

Public blockchain data is also not the same as bank AML. Elliptic provides a valuable temporal graph task, but unknown labels, anonymized features, and public-chain behavior limit direct operational interpretation. Elliptic2 is modern and highly relevant, but the current local evidence does not satisfy the stronger reference-parity conditions needed for a performance contribution against RevClassifyDS.

The deterministic system checks are not a substitute for a human usability study. They show that artifacts, redactions, interpretation gates, and recovery surfaces are present and internally consistent. They do not measure analyst time, production incident rates, organizational adoption, or real investigation quality. Future work should test stronger release budgets, repeated runs, private or partner-approved holdouts, same-queue incumbent comparisons, and graph-native families under the same evidence-cell discipline.

The system is intentionally local-first, which creates a tradeoff. Privacy and provenance improve because raw rows stay local, but external reviewers cannot rerun licensed or private data without obtaining it themselves. The paper handles that by publishing commands, hashes where allowed, generated artifacts, and claim boundaries, but a fully independent reproduction of every heavy benchmark still depends on legal access to the source datasets.

9 Reproducibility

The repository is larger than this AML paper. Relaytic is the general local-first inference lab and public package; Relaytic-AML is the focused AML edition used here for the manuscript. A reader should start with the README and this paper. Development-control files record the build history, but they are not required to understand the paper claims.

Table 8. Reproducibility contract.

Component	Command	Expected output	Environment or data dependency	Hash or seed record
Paper artifacts	paper-release; paper-arxiv-source	Markdown draft, arXiv main.tex, vector figures	Python >=3.10; CI matrix covers 3.10 and 3.11; full extra installs numpy, pandas, scikit-learn, matplotlib	deterministic generators; source hashes in manifest
PaySim benchmark	release-safety paysim-competitive –budget-tier competitive –run-optional –format json	PaySim selected PR-AUC cell	local Kaggle PaySim file required	sha256 prefix 16910f90577b
Elliptic benchmark	release-safety graph-baselines –budget-tier competitive –run-optional –format json	Elliptic selected PR-AUC cell	local Kaggle Elliptic files required	sha256 prefix 93e2e7b2405c plus 2 files
System evaluation	release-safety paper-system-eval –format json	task, handoff, recovery, and claim-gate reports	repo-local deterministic fixtures	all required system-evaluation tasks pass in current evidence pack

Minimal regeneration commands are shown below.

Windows PowerShell:

```
py -3.11 -m pip install -e ".[full]"
py -3.11 -m relaytic.ui.cli release-safety paper-system-eval --format json
py -3.11 -m relaytic.ui.cli release-safety paper-release --format json
py -3.11 -m relaytic.ui.cli release-safety paper-arxiv-source --format json
py -3.11 -m pytest tests/test_paper_track_p13.py tests/test_paper_track_p14.py tests/test_paper_tra
```

macOS/Linux:

```
python3 -m pip install -e ".[full]"
python3 -m relaytic.ui.cli release-safety paper-system-eval --format json
python3 -m relaytic.ui.cli release-safety paper-release --format json
python3 -m relaytic.ui.cli release-safety paper-arxiv-source --format json
python3 -m pytest tests/test_paper_track_p13.py tests/test_paper_track_p14.py tests/test_paper_trac
```

Raw benchmark data is not committed. PaySim and Elliptic require local downloads and are referenced through registry artifacts, split reports, hashes, and command ledgers. Elliptic2 remains context-only in this paper because the stronger reference-parity conditions are not satisfied locally. Clean clones can reproduce the paper-generation checks and repo-local public fixtures; full benchmark regeneration requires the locally licensed datasets named in the README.

10 Use of AI Assistance

Large language model tools assisted with drafting, editing, repository inspection, consistency checks, and implementation work around the paper artifacts. They are not authors. The evidence cells, benchmark outputs, source code, figures, tables, limitations, and final interpretation remain the author's responsibility.

11 Conclusion

Relaytic-AML shows how an agent-assisted AML evaluation lab can be built around local evidence rather than conversational memory. The system keeps data posture, temporal and graph split validity, leakage controls, model budgets, review-budget operating points, rowless handoff, and public claims inside one artifact record. The PaySim, Elliptic, and Elliptic2 rows are useful because they demonstrate that architecture under realistic forms of pressure, including rare events, graph provenance, modern benchmark context, and governed interpretation.

The strongest claim supported today is architectural: Relaytic-AML can make AML experiments easier to inspect, easier to challenge, safer to hand off to another agent, and harder to overstate. That is the useful substrate on which stronger detector studies, private holdouts, incumbent comparisons, and graph-native budgets can be built.

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